

Basic Principles of Medicine 1
Module : Foundations to Medicine
FTM 16

Title: Patterns of Inheritance – III
Unexpected patterns

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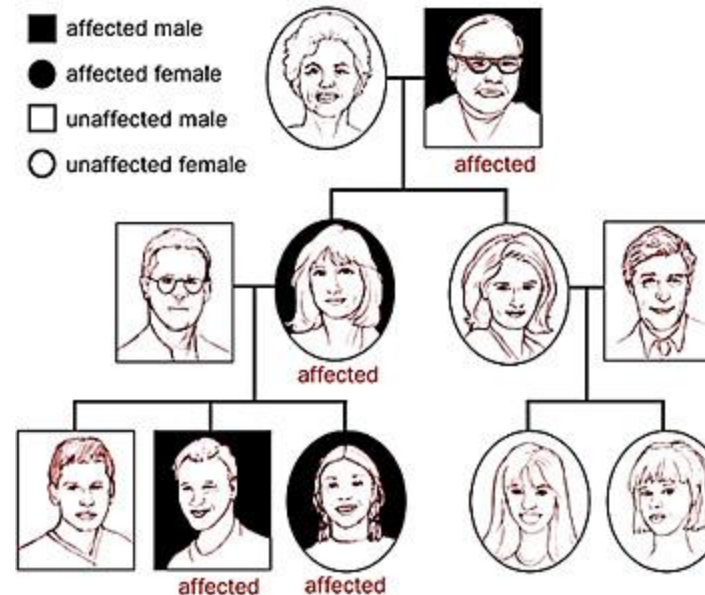
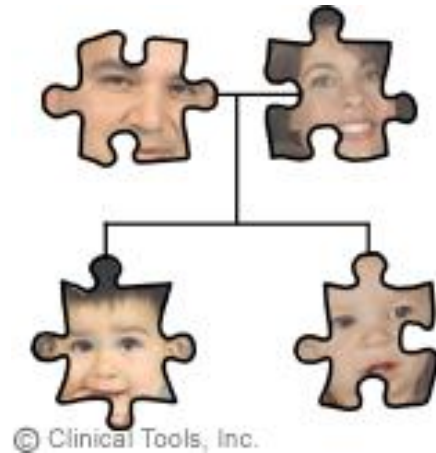
Recommended Reading

Human Genetics and Genomics: Korf & Irons 4th
Edition; Chapter 3

Objectives

SOM.MKIII.BPM1.1.FTM.3.GNET.0059	Comprehend how the classic pattern of inheritance may be disrupted by incomplete penetrance, variable expressivity and variable age-of-onset. Be able to identify these concepts from clinical scenarios and pedigrees
SOM.MKIII.BPM1.1.FTM.3.GNET.0060	Describe examples of genetic abnormalities that may result in a non-Mendelian pattern of inheritance
SOM.MKIII.BPM1.1.FTM.3.GNET.0061	Discuss disorders due to mutations in mitochondrial genes: Leber hereditary optic neuropathy (LHON), MELAS, MERRF
SOM.MKIII.BPM1.1.FTM.3.GNET.0062	Identify the phenomenon of heteroplasmy in mitochondrial disorders and indicate its genetic basis
SOM.MKIII.BPM1.1.FTM.3.GNET.0063	Predict the effect of imprinting in the transmission of genetic disease
SOM.MKIII.BPM1.1.FTM.3.GNET.0064	Identify imprinting defects (Prader Willi syndrome, Angelman syndrome)
SOM.MKIII.BPM1.1.FTM.3.GNET.0065	Explain the phenomenon of uniparental disomy and microdeletion in Prader-Willi and Angelman syndrome
SOM.MKIII.BPM1.1.FTM.3.GNET.0066	Analyze how triplet repeat expansion mutations result in anticipation; discuss its genetic basis and identify it from clinical scenarios and pedigrees and how this may be thought about as a non-Mendelian aspect to the inheritance of the disorder (Huntington disease, myotonic dystrophy, Fragile X syndrome)
SOM.MKIII.BPM1.1.FTM.3.GNET.0067	Explain digenic inheritance using retinitis pigmentosa as an example
SOM.MKIII.BPM1.1.FTM.3.GNET.0068	Determine a strategy for identifying Non-Mendelian disorders, investigating them and providing appropriate genetic counselling for families
SOM.MKIII.BPM1.1.FTM.3.GNET.0110	Demonstrate how genomic imprinting would appear in a human pedigree when either a male or female passes a mutation to offspring

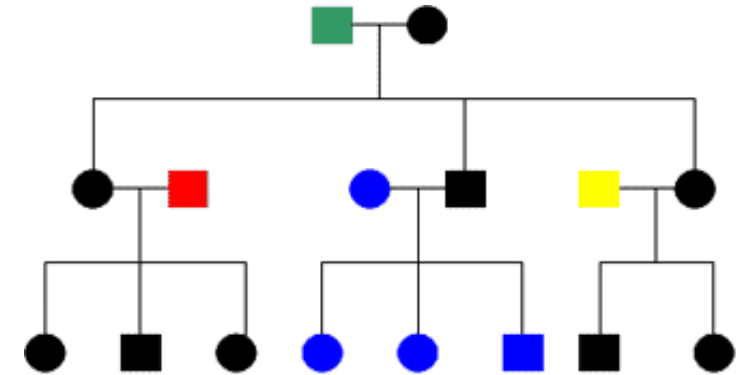
Patterns of inheritance III



An important thematic objective of these lectures is to encourage the student to learn how to create a three generation family pedigree from a patient history. From this, many genetic phenomena may be recognized

Mitochondrial Inheritance

- Mitochondria are inherited from the mother
- All offspring of an affected female are affected
- Only females transmit the disorder
- Affected father does not transmit the disorder to his children
- Both male and female children of an affected female are affected



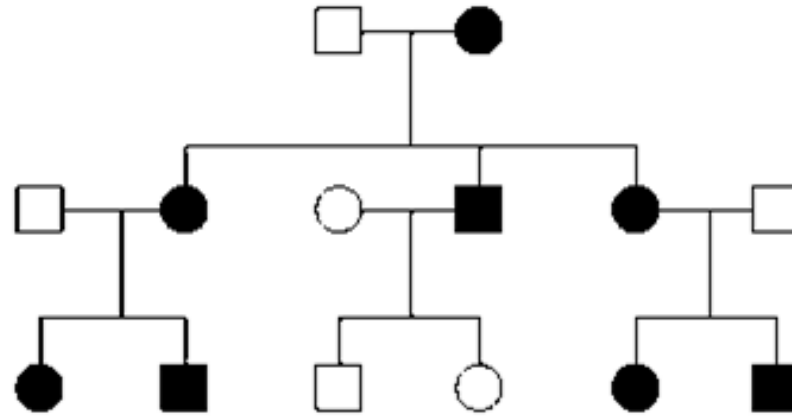
http://www.bbc.com/news/magazine-28986843?ocid=socialflow_facebook

The girl with three biological parents

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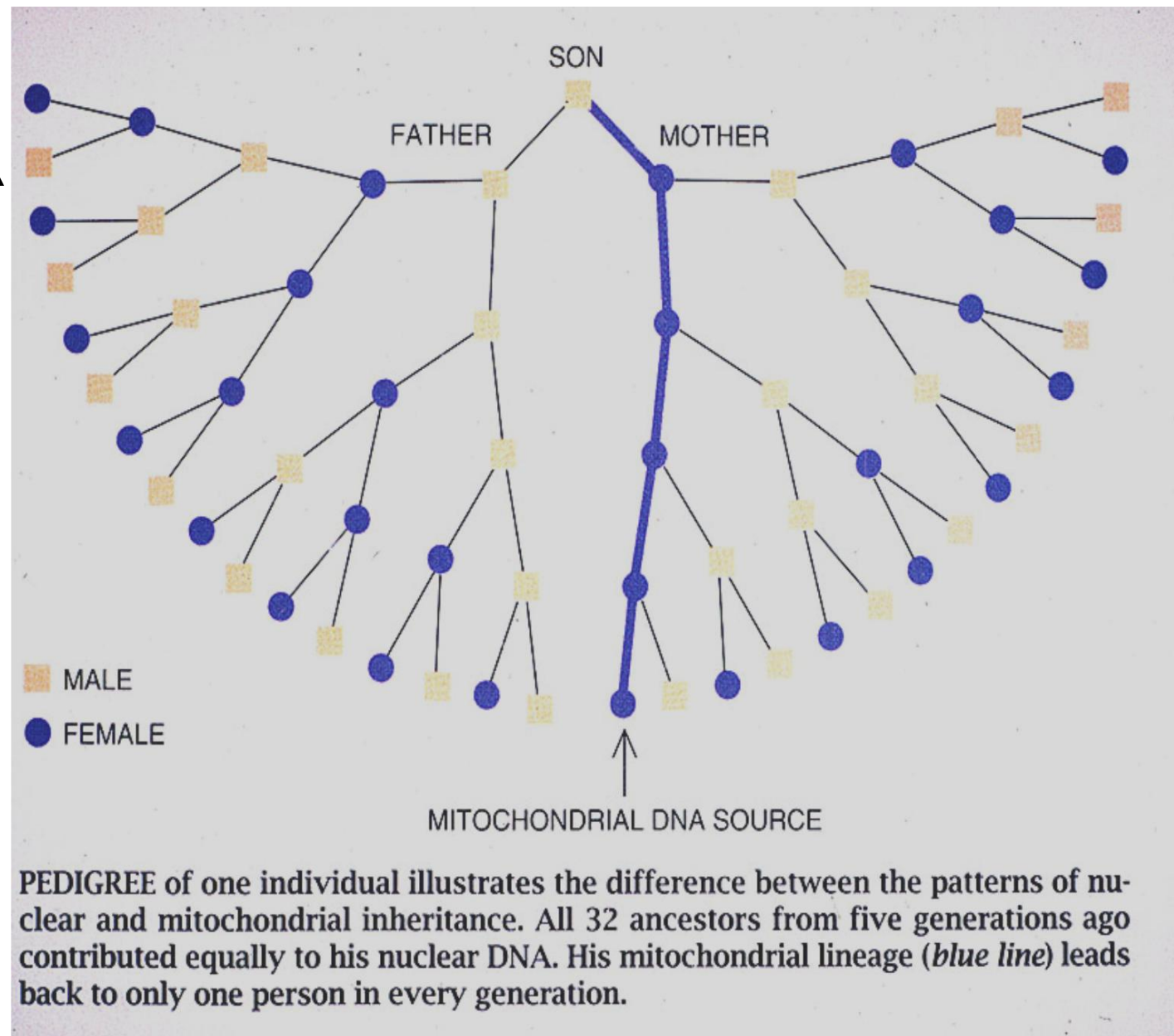
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Mitochondrial Inheritance



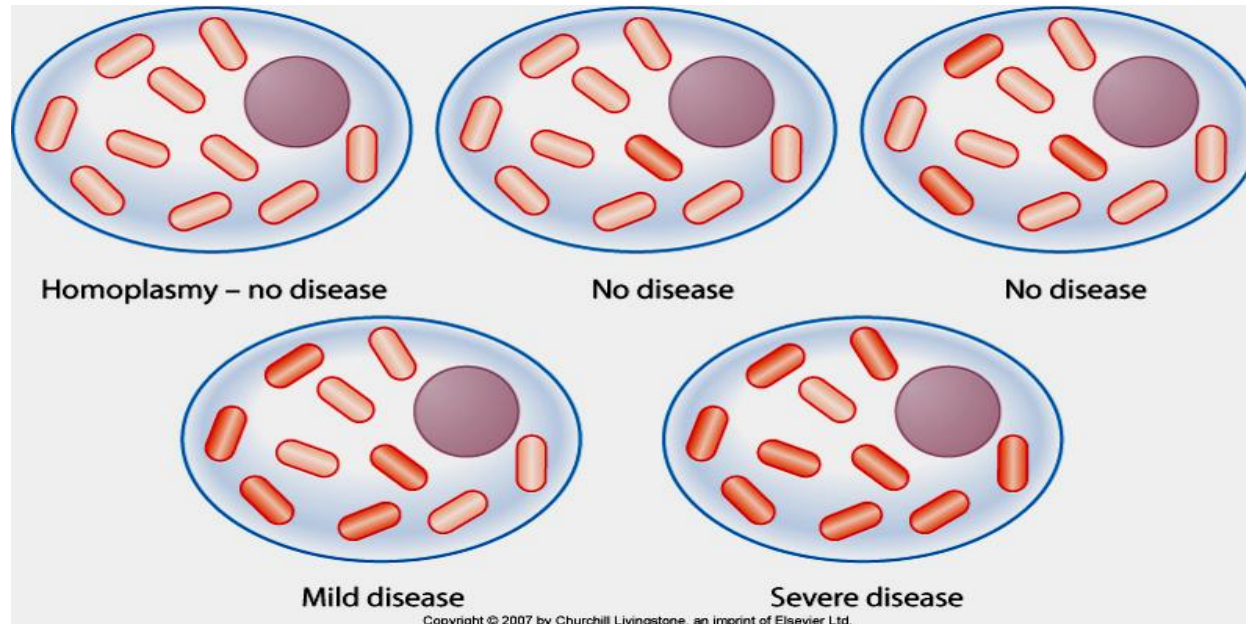
- **ALL offspring** of an affected female are affected
- Only females transmit the disease
- Affected males do NOT transmit the disease
- Both male and female children of an affected female are affected

*Direction of
inheritance*



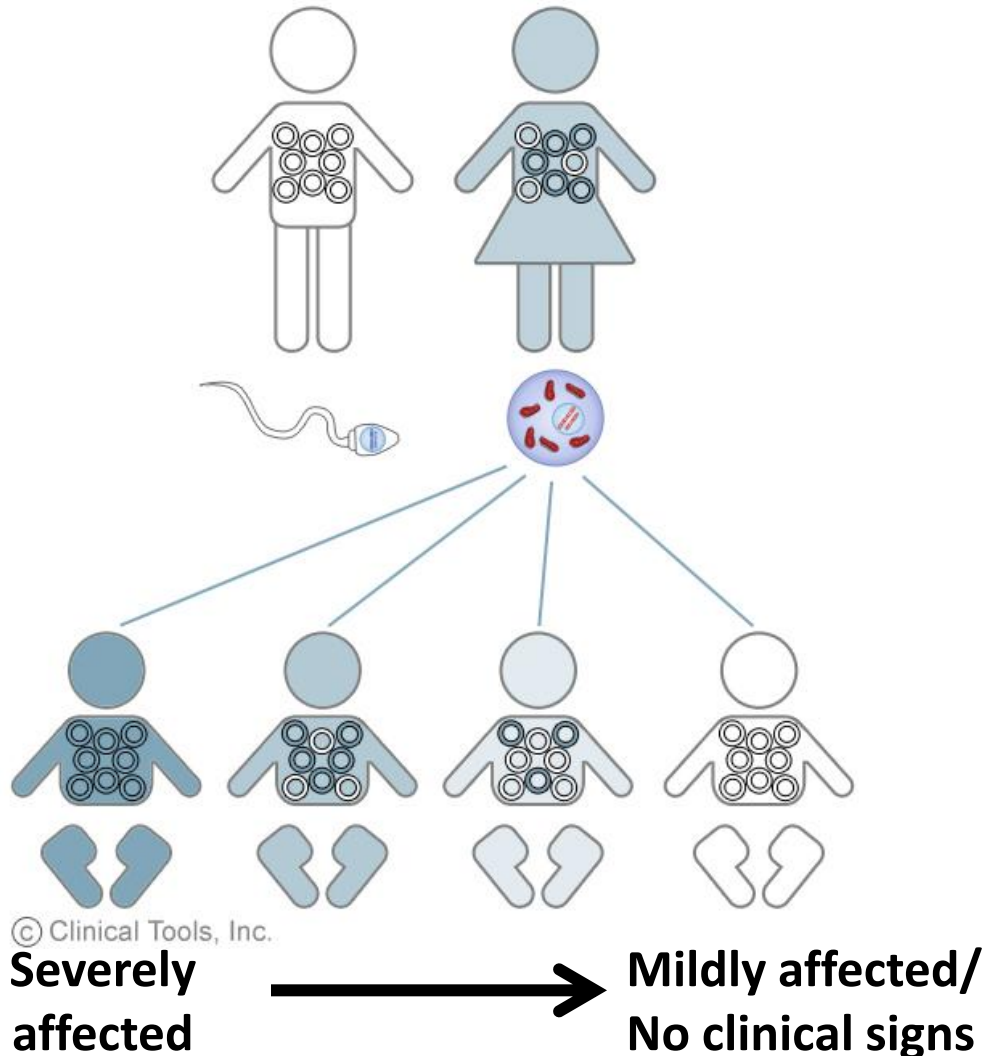
Variable expression in mitochondrial disorders

- The severity of mitochondrial disorders can vary in individuals with the disorder
- This is due to the phenomenon of **heteroplasmy**
- The severity of the disorder depends on the number of mitochondria that have the mutant gene



Heteroplasmy in mitochondrial disorders

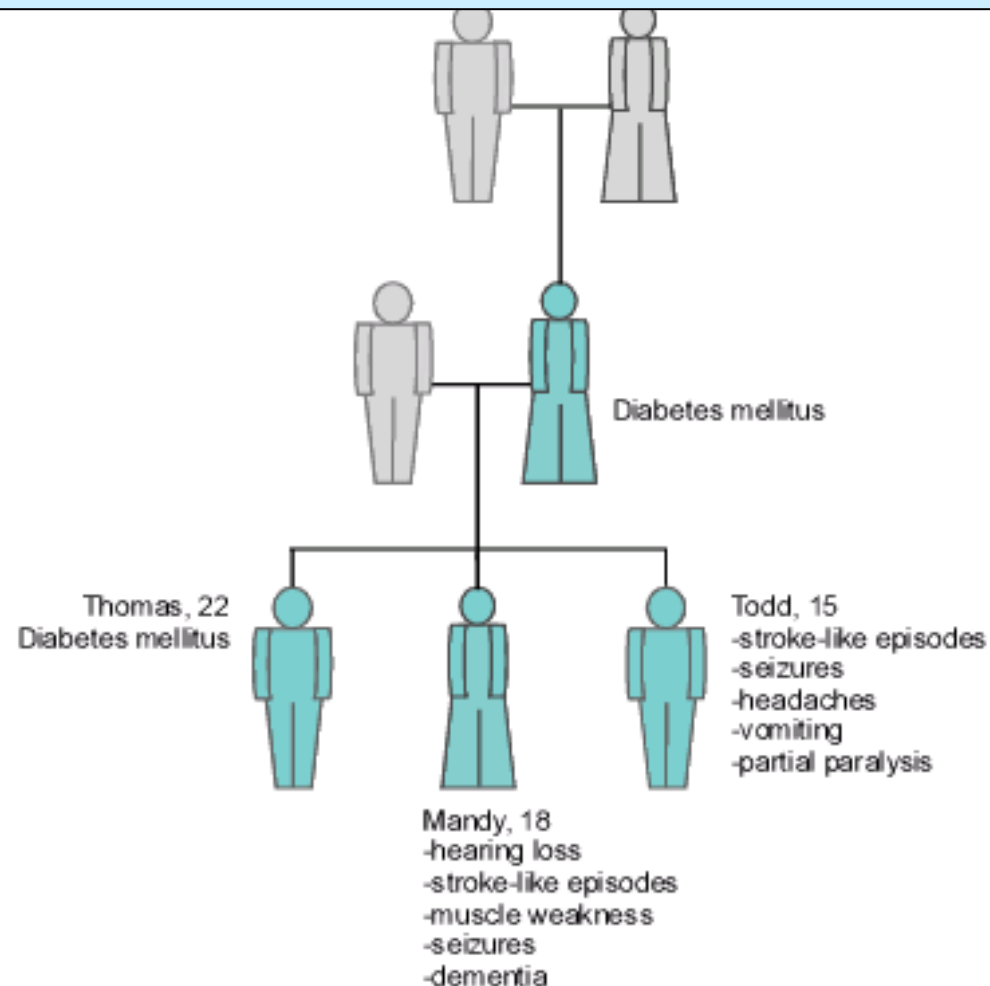
<http://www.bioedonline.org/videos/content-presentations/human-organism/mitochondrial-genetics-and-diseases/>



- Mitochondrial DNA segregates passively when a cell divides
- Unequal distribution of mutant and non-mutant mitochondrial DNA, results in the phenomenon of heteroplasmy in mitochondrial disorders
- Explains the variable expression of the disease in the offspring.

Case Example (mitochondrial inheritance and heteroplasmy): MELAS

Thomas is seen for genetic counseling because a diagnosis of MELAS, an acronym which describes the associated phenotype - Mitochondrial Encephalomyopathy, Lactic Acidosis, and Stroke-like episodes, has been made in his sister and brother



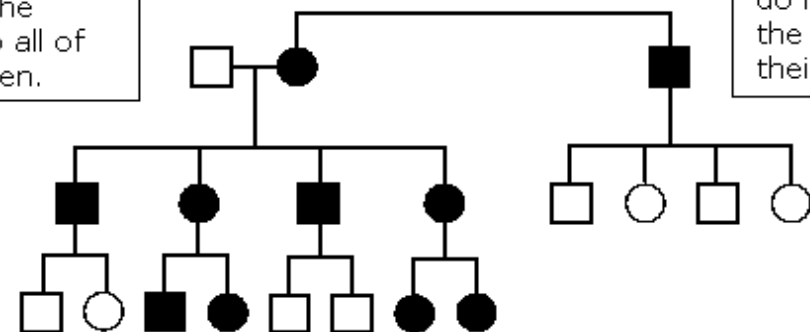
What is the risk of the disorder being present in Thomas's children?

Key

- Female, unaffected
- Female, affected
- Male, unaffected
- Male, affected

Affected females transmit the disease to all of their children.

Affected males do not transmit the disease to their children.

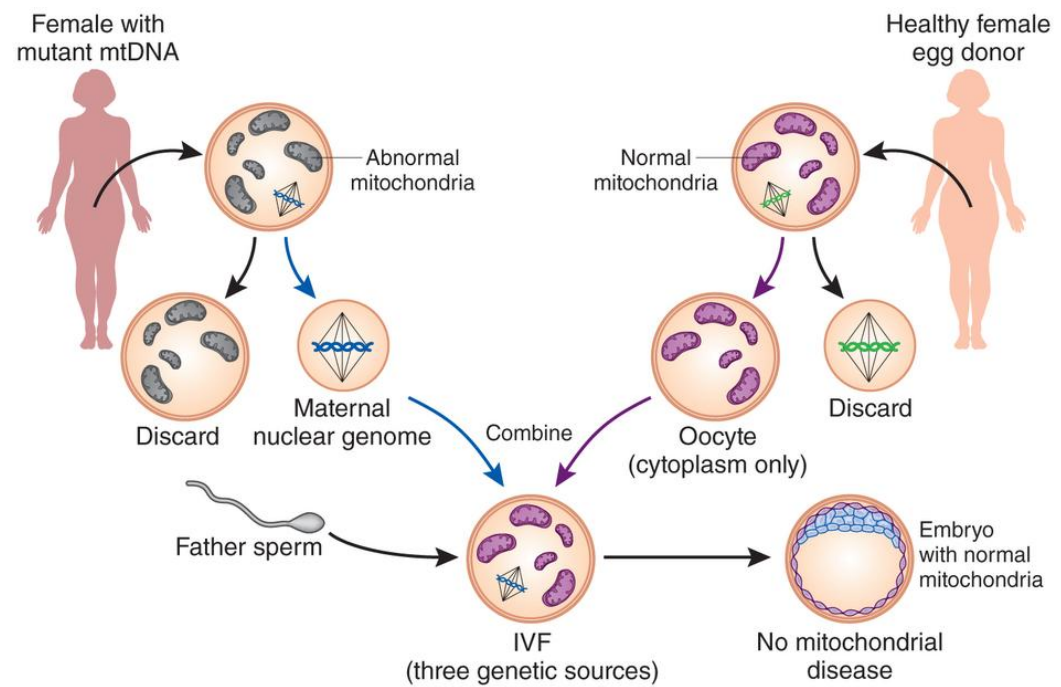
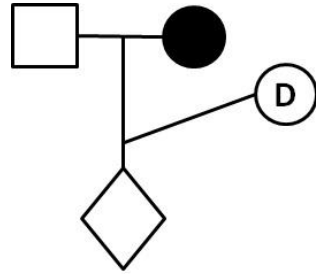


Read more about mitochondrial inheritance and MELAS in Korf and Irons 4th edition p57-61

Mitochondrial disorders

<http://www.bbc.com/news/magazine-28986843?ocid=socialflow>
[facebook](#): The girl with three biological parents

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- Mitochondrial mutations generally affect multiple organ systems
- Leber hereditary optic neuropathy
 - Manifests as progressive blindness around 20-30 years
- MELAS: Mitochondrial encephalopathy, lactic acidosis, and stroke like episodes
- Myoclonic epilepsy with ragged red muscle fibres (MERRF)
- When a female is affected with a severe mitochondrial disorder, one of the options offered is artificial reproduction technology
- The mom's nuclear DNA is fused to the donor's ovum (enucleated) and fertilized with the dad's sperm cells

Disorders with non-Mendelian/ unexpected inheritance patterns

- Digenic disorders
 - Retinitis pigmentosa
- **Imprinting**
 - Prader Willi syndrome
 - Angelman syndrome

} Parent of origin effect
- **Triplet repeat disorders with anticipation (Mendelian – but mutation is unstable)**
 - Huntington disease
 - Myotonic dystrophy
 - Fragile X syndrome (X-linked)

} Autosomal dominant

Digenic disorders

- Mutations in two genes (A, B) are additive and necessary to produce the disorder

AABB **AABb** **AaBB** are all normal

AaBb shows the disease phenotype (Heterozygotes for gene A and B)

(individuals with mutations in both the genes demonstrate the phenotype)

- One form of retinitis pigmentosa
 - a disease of progressive visual impairment
 - a result of a mutation in two independent genetic loci (ROM1 and peripherin)
 - Heterozygotes for gene A and gene B have retinitis pigmentosa (**AaBb**)

Two mutant loci work together to produce an effect which is not seen without the other!

(AABB)

ROM1 (gene A)

Peripherin (gene B)

Contribute to photoreceptors (rods)



~~*ROM1 (gene a)*~~

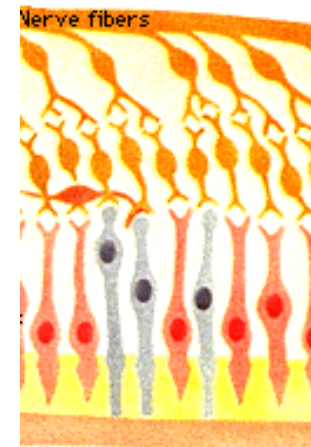
Peripherin (gene B) (AaBB)

OR

ROM1 (gene A)

~~*Peripherin (gene b) (AABb)*~~

Normal
photoreceptors



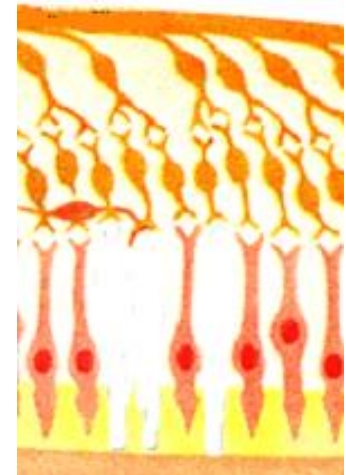
but

~~*ROM1 (gene a)*~~

~~*Peripherin (gene b)*~~

(AaBb)

Retinitis pigmentosa and
degenerating rods

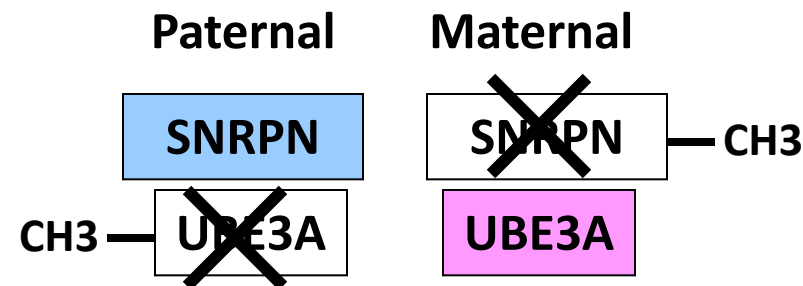


Imprinting (Parent of origin effects)

- Some genes are active only when transmitted by mother or father
- Imprinting involves **methylation** of specific loci (epigenetic change) and **silencing** of the gene
- **Normal imprinting pattern on chromosome 15:**
 - SNRPN gene is silenced by methylation on maternal ch15, but **UBE3A** is active.
 - **UBE3A** is active on the **maternal** chromosome 15
 - On paternal ch15, **SNRPN** is active and UBE3A is inactive (silenced by methylation).
 - **SNRPN** is active on the **paternal** chromosome 15

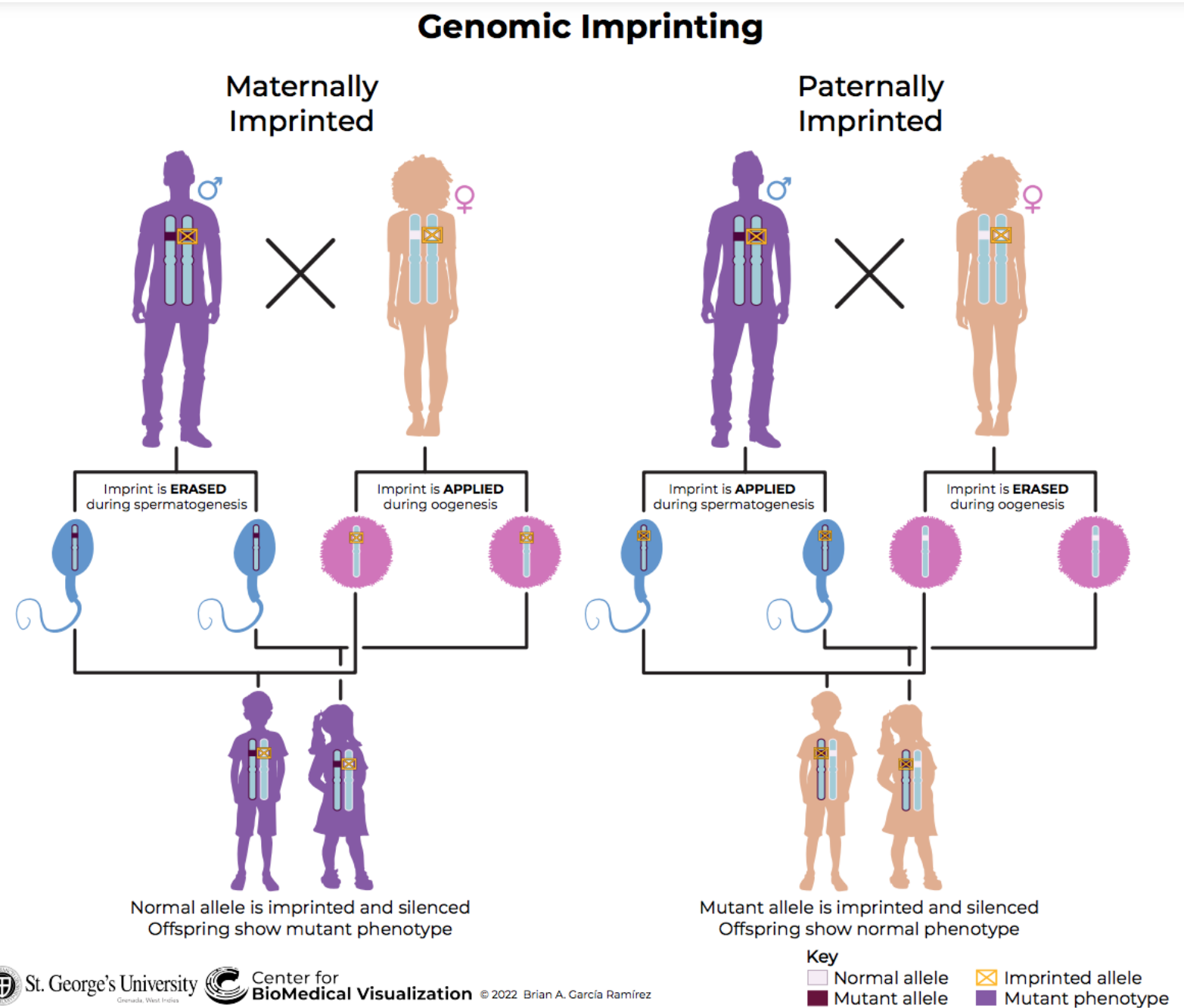
Normal imprinting pattern on chromosome 15

<http://www.genetics.edu.au/publications-and-resources/facts-sheets/fact-sheet-14-epigenetics>



Genomic imprinting can be loosely defined as the gamete-of-origin dependent modification of phenotype.

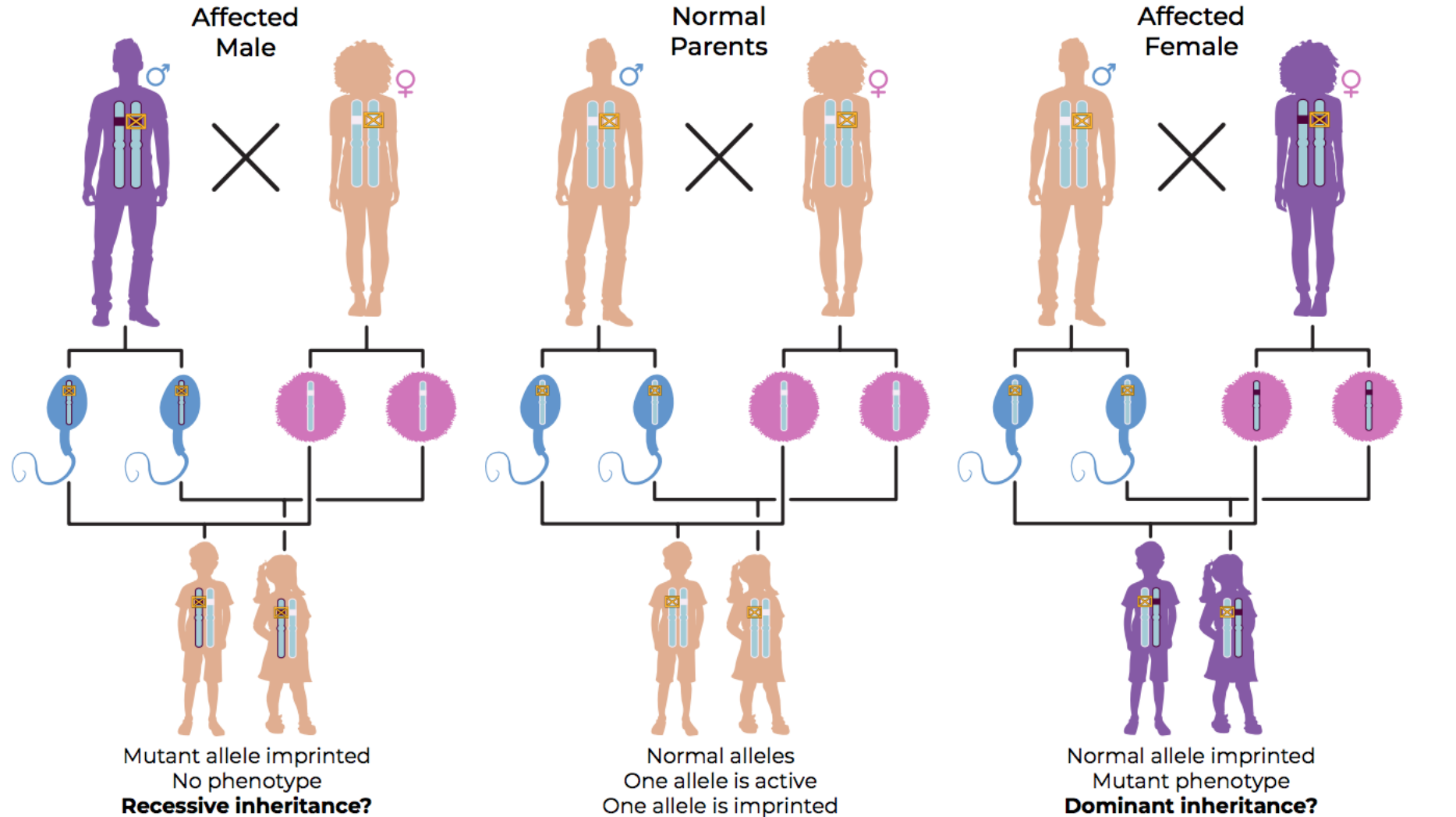
That is, the parental origin of a particular allele will determine if that that particular allele is expressed - the imprinted allele being inactivated.



Genomic imprinting can make mutant alleles appear to behave as both dominant and recessive.

Example, **hypothetical paternally imprinted gene.**

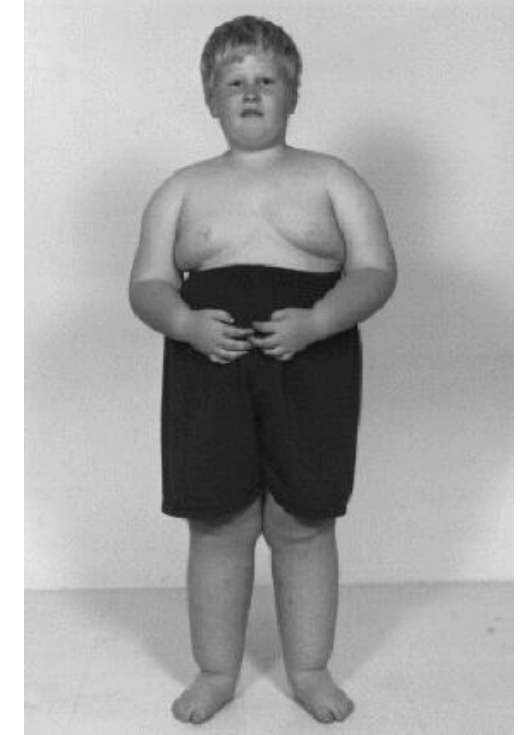
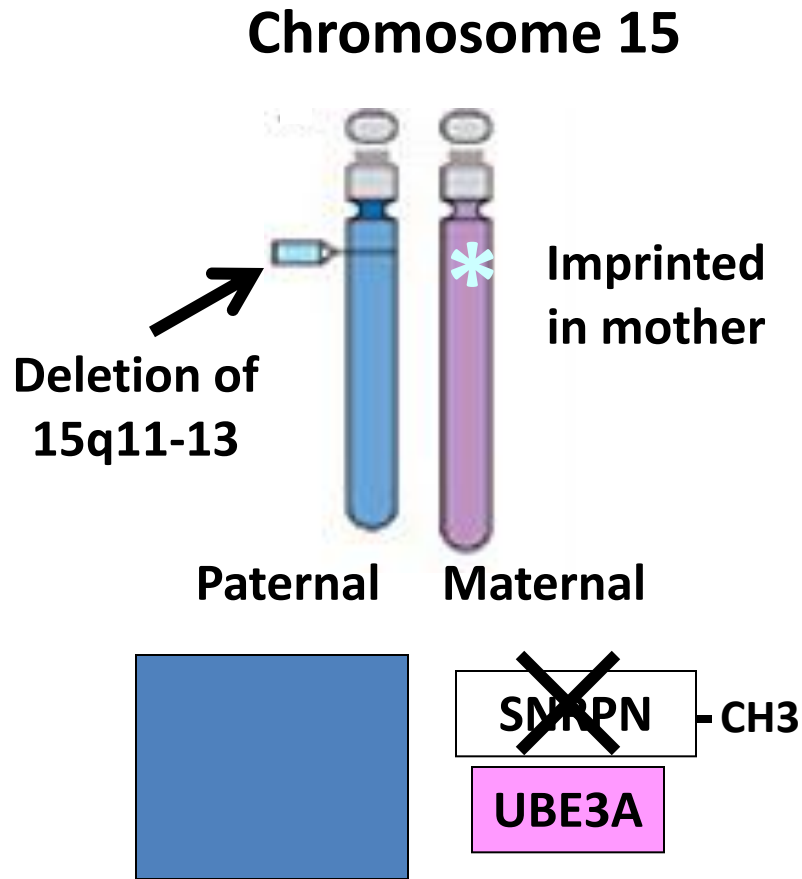
Hypothetical Case of Paternally imprinted gene



Prader Willi Syndrome

- Prader Willi syndrome may be caused by:
 - Microdeletion of paternal chromosome 15 (70% of patients)
 - Maternal uniparental disomy of chromosome 15
- Prader Willi syndrome due to microdeletion of **Paternal** 15q11-13 (absence of SNRPN)
- Microdeletion of this region in **paternal** chromosome → Prader Willi syndrome (about 70% of cases)
- Microdeletion can be detected by FISH using specific probes against the region or by array CGH (CMA)
- Children are usually obese, have mental and developmental delay and underdeveloped genitalia
- Hypotonia in infancy, failure to thrive

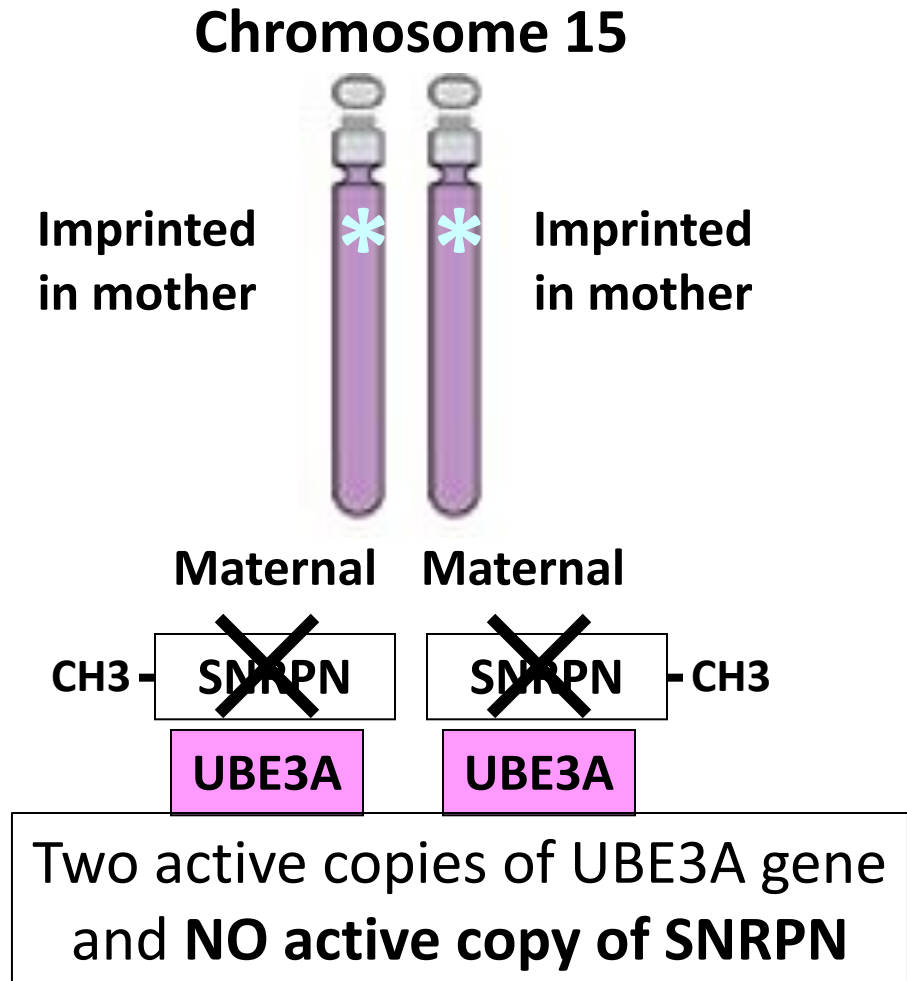
Prader Willi Syndrome - due to microdeletion



UBE3A gene active and
NO active copy of SNRPN

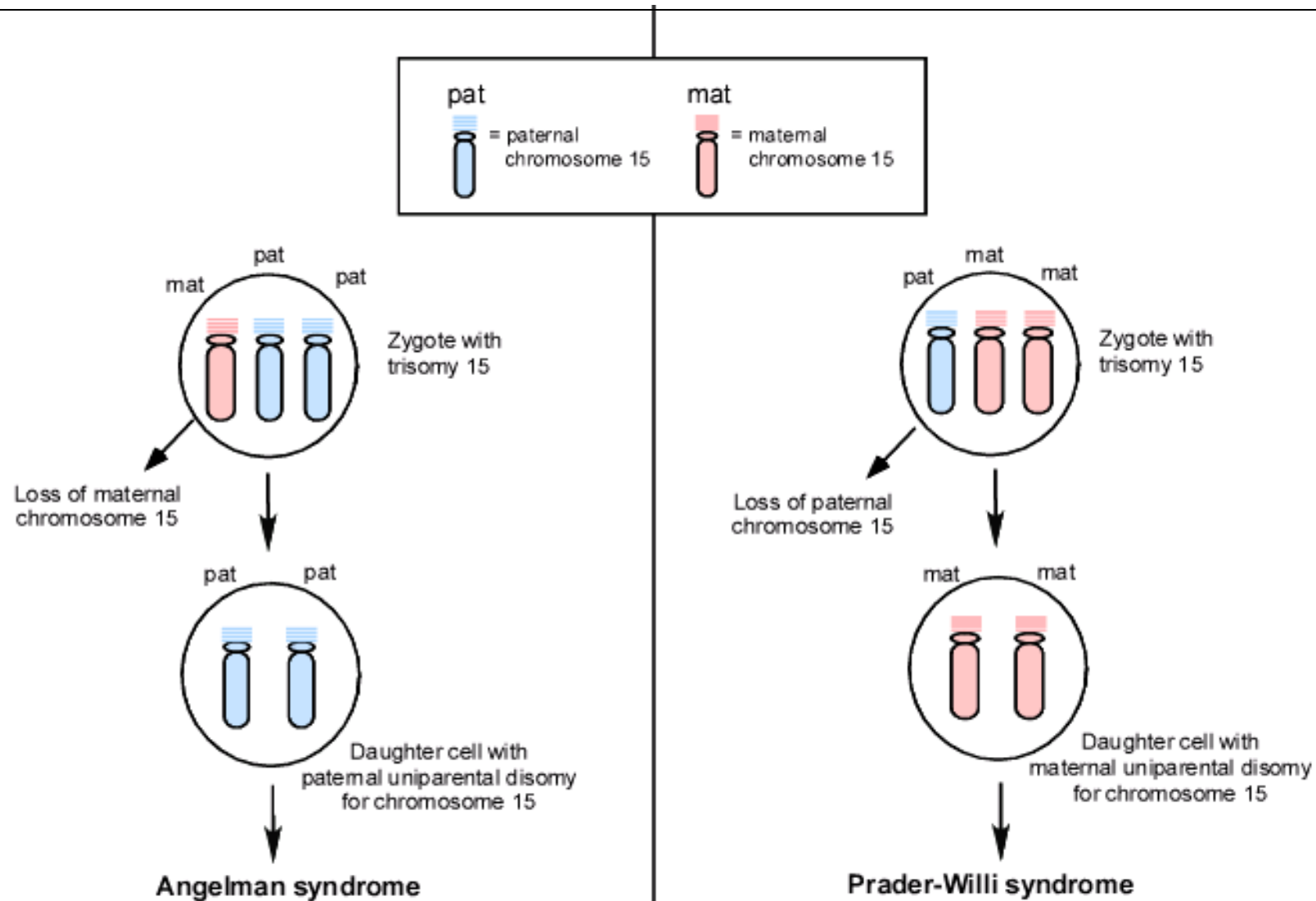
In children with microdeletion of the region on paternal chromosome 15, there is production of **UBE3A** protein from **maternal gene**, but **SNRPN** that is normally produced from the paternal chromosome 15 is absent, resulting in the phenotypic manifestations

Prader Willi Syndrome - Uniparental disomy



- Uniparental disomy is a condition when the child has two copies of the **maternal chromosome 15** (absence of paternal chromosome 15)
- The region 15q11-13 is normally inactive in the maternal chromosome due to imprinting (methylation)
- Such children have two active copies of UBE3A and **NO active copies of SNRPN gene**
- Can be detected by methylation analysis—there are two methylated copies of ch15

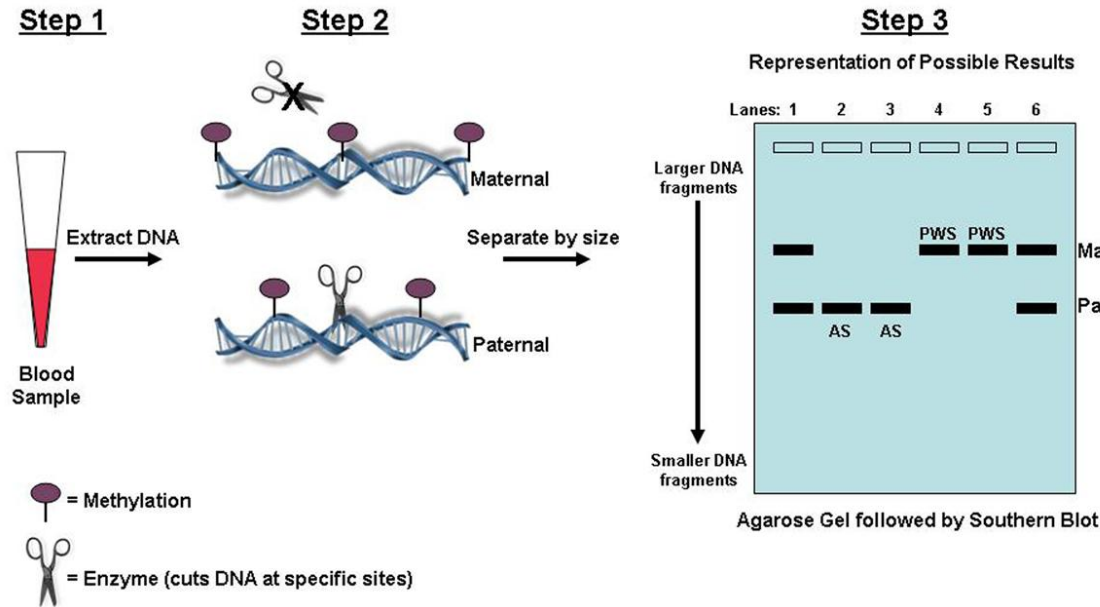
Uniparental disomy- phenomenon of trisomy rescue



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Methylation analysis

The probe being used in this example is for the *Snrpn* gene

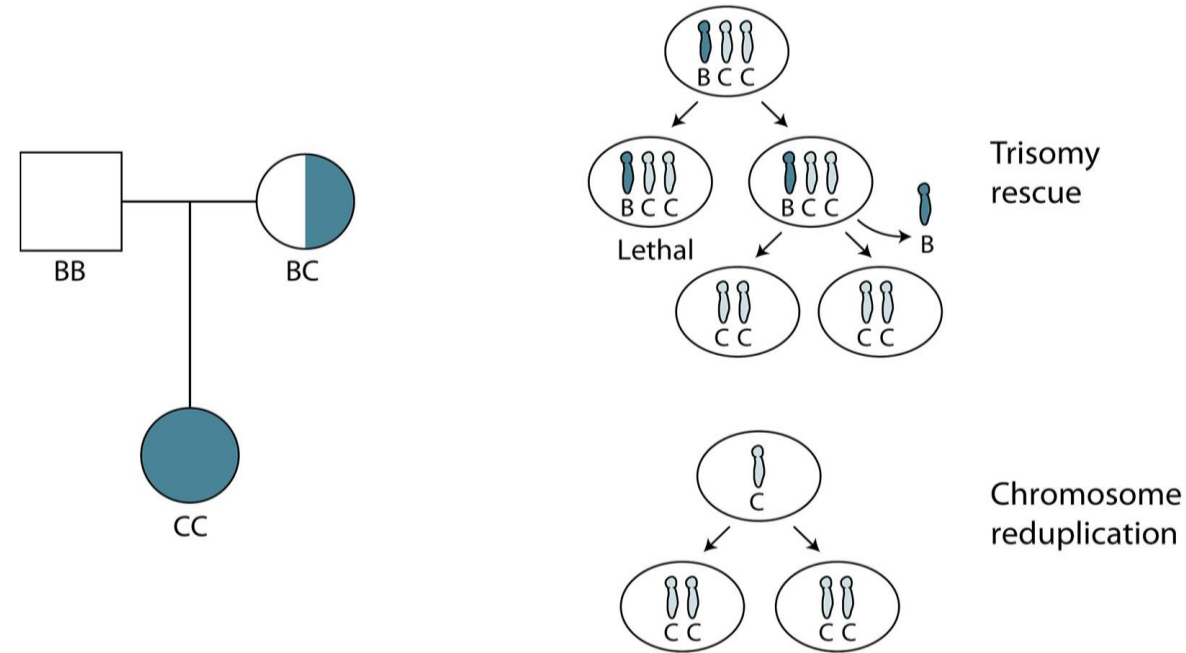


The maternal copy of *Snrpn* is imprinted, and methylated. The restriction enzyme is unable to cut this DNA segment resulting in a large DNA fragment.

The paternal copy of *Snrpn* is not imprinted, and is unmethylated. The restriction enzyme is able to cut this DNA segment resulting in a small DNA fragment.

- Methylation sensitive restriction enzyme analysis uses the differential methylation patterns of chromosome 15 (maternal and paternal)
- The restriction enzyme does not cleave methylated maternal DNA; Paternal chromosome 15 at this region is cleaved by the enzyme
- May be exploited by Southern blot, PCR and DNA sequencing
- Methylation pattern analysis is most useful when the children are very young and have not developed classical phenotypes of the disorders

Uniparental disomy in Prader-Willi syndrome



Human Genetics and Genomics, Fourth Edition. Bruce R. Korf and Mira B. Irons.
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Figure 6.27 A child is homozygous for allele C, inherited from the mother, due to the phenomenon of uniparental disomy. This occurs by loss of the paternal chromosome, containing allele B, from a trisomic zygote.

Polymorphic marker analysis can be used to differentiate the cause of Prader-Willi syndrome (due to microdeletion of paternal ch15 or due to uniparental disomy of maternal ch15)

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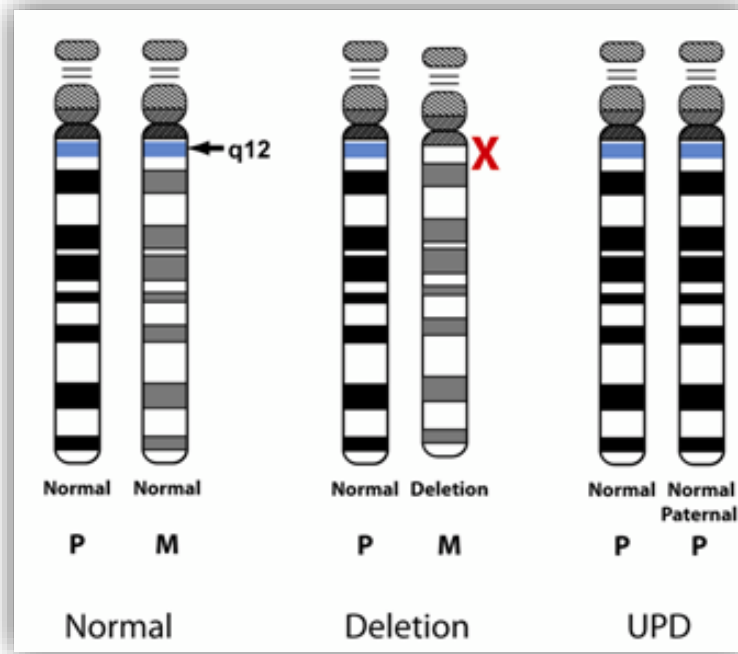


Angelman syndrome

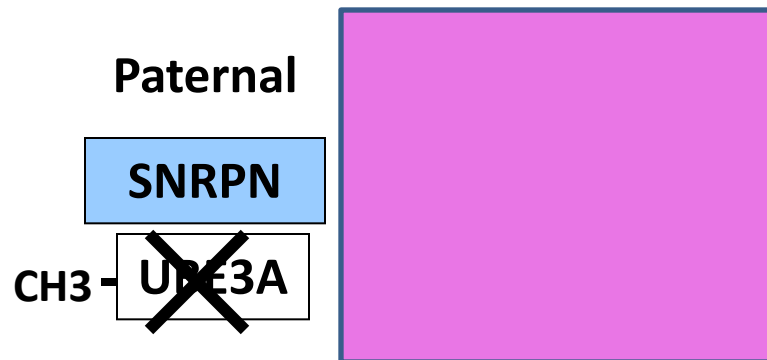
- Angelman syndrome can be caused by:
 - Deletion of **maternal 15q11-13 (absence of active UBE3A gene)**
 - **Uniparental disomy** of paternal chromosome 15 (two copies of active SNRPN and **absence of UBE3A gene**)
- “Happy Puppet syndrome”
- Happy disposition, laugh inappropriately
- **Severe intellectual disability, seizures**
- **Puppet like posture of limbs**



Angelman syndrome



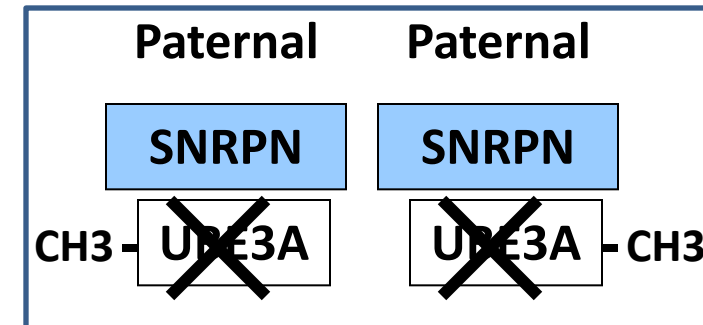
Maternal deletion of chr 15 region



SNRPN gene active and **NO active copy of UBE3A**

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Paternal uniparental disomy of chr 15

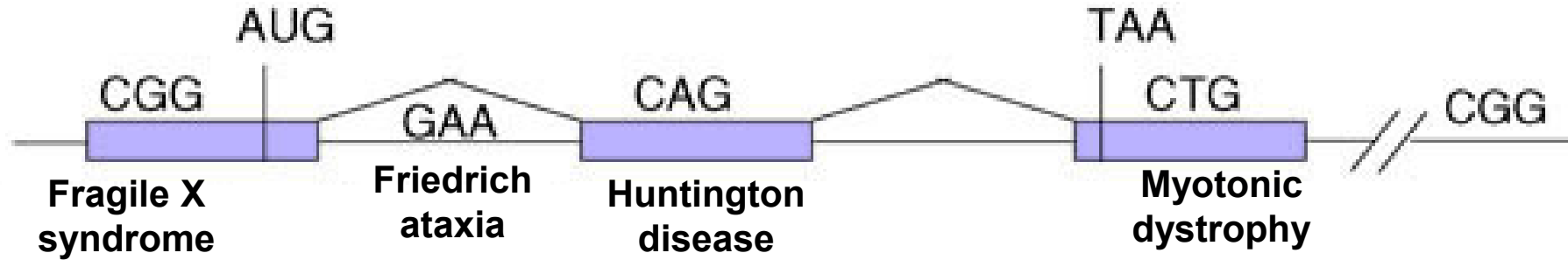


Two active copies of SNRPN gene and **NO active copy of UBE3A**

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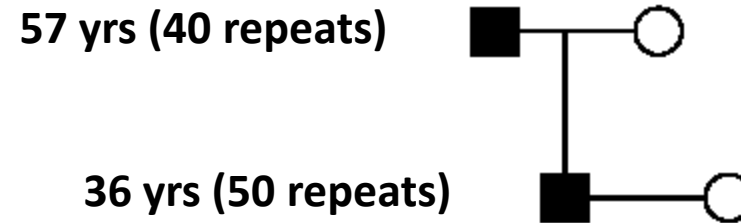
Four classes of triplet repeat disorders

(based on location of the repeat)



- The triplet repeat may be present
 - At the **promoter region** of the gene, resulting in reduced expression of the gene (**CGG repeat** in Fragile X syndrome)
 - In an **intron**, resulting in formation of heterochromatin (Friedrich ataxia)
 - In the **coding region of the gene**, resulting in a polyglutamine expansion in the protein (Huntington disease: **CAG** codes for glutamine)
 - At the **3' end (3' UTR) of the gene** (myotonic dystrophy)

Anticipation



- Individuals in the recent generations of a pedigree develop disease at an earlier age and generally with greater severity
- **Huntington disease**
 - Triplet repeat expansion: CAG repeat in the exon
 - There is a **polyglutamine tract** in the huntingtin protein
- Greater the number of repeats, earlier is the age of onset of symptoms and more severe is the disorder
- Repeats are **unstable**, and repeat size influences disease severity and age of onset – explains phenomena called **anticipation**

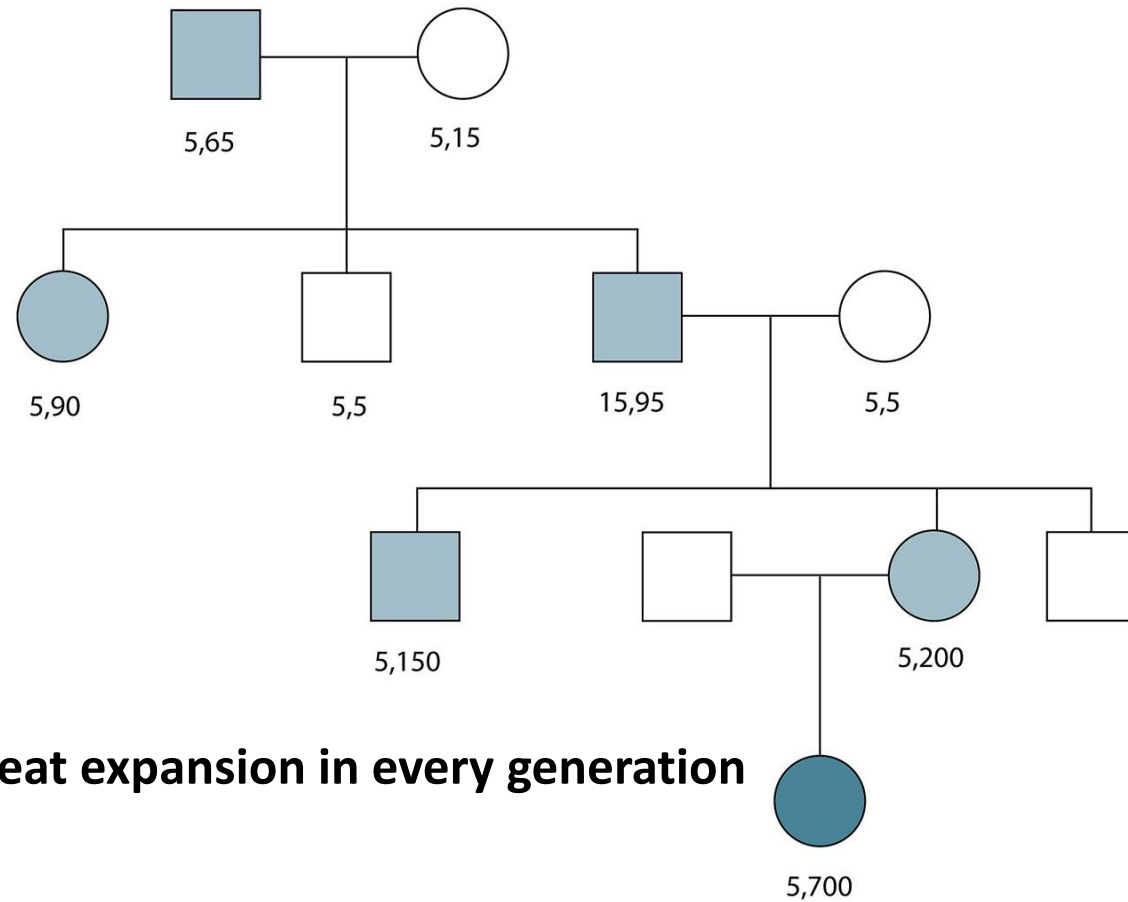
Anticipation

- The larger the repeat, the greater is the chance for it to be **unstable** and it tends to expand from one generation to the next
- Anticipation is observed in the triplet repeat expansion disorders:
 - In **Huntington disease** (autosomal dominant disorder), is generally more severe if transmitted from the father
 - **Myotonic dystrophy** (autosomal dominant disorder)- expansion takes place in the mother
 - **Fragile X syndrome** (X linked disorder)- repeat expansion in the mother

Anticipation in myotonic dystrophy



Increasing severity
from one generation
to the next



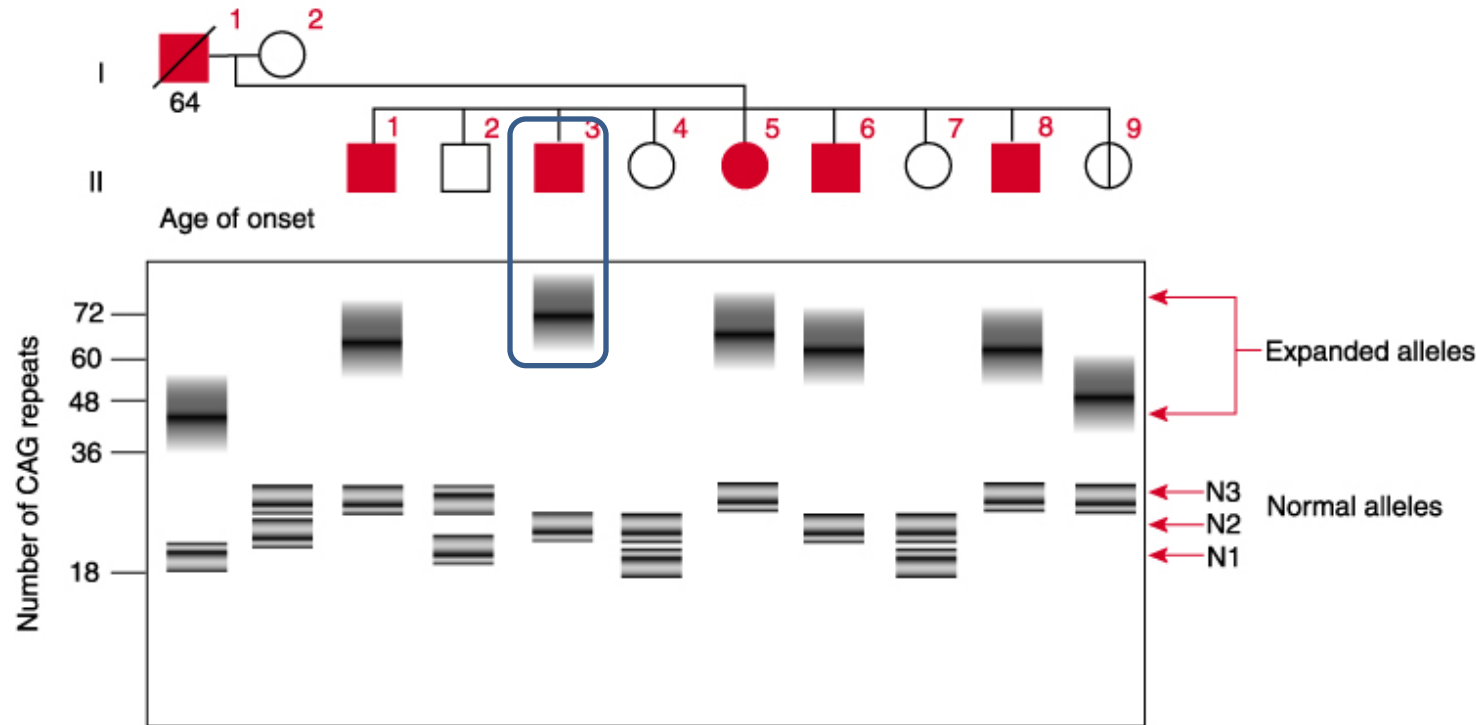
Note the repeat expansion in every generation

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Figure 3.28 Anticipation in myotonic dystrophy. The number of CTG repeats is indicated below each symbol. Each individual has one normal allele of five repeats. In the first generation, the affected individual has a 65-repeat allele that gives rise to mild myotonic dystrophy. This increases to 95 repeats in his affected son, then to 200 repeats in his affected daughter, and finally to 700 repeats in the severely affected child in the last generation.

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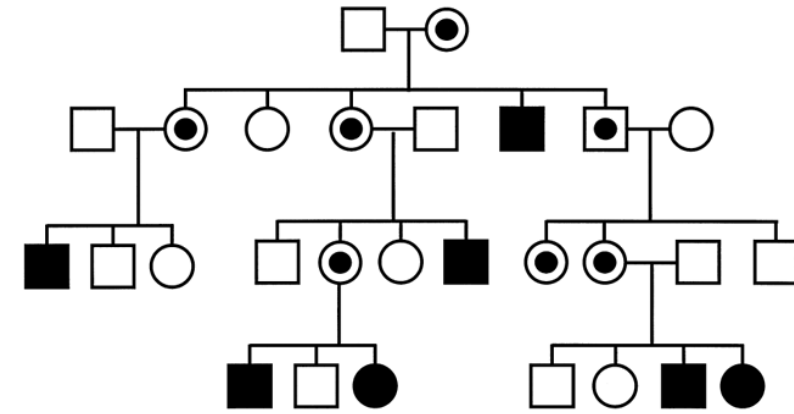
In a family with Huntington disease, the Southern blot analysis of chromosome 4 is depicted below. In which individual in the family would you suspect an earlier age of onset of the symptoms? Explain



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What may be the reason for the phenotype of II-9?

Fragile X syndrome



- A mother brings her 3-year-old twin **boys**. One of them had some physical characteristics associated with babies with fragile X - poor muscle tone, a sunken chest and an elongated head. And, he **couldn't speak**; a typical fragile X child will begin developing language skills at 4 or 5 years. His other symptoms include difficulty maintaining eye contact, frequent hand slapping when excited.
- The second twin experiences the **anxiety** and **temper tantrums** considered hallmarks of this condition.
- A 6-year-old sister has mild learning disability. The mother is normal.

Fragile X syndrome

- Triplet repeat expansion (CGG repeat) on the X chromosome.
- The triplet repeat is present at 5' end (**promoter region**) of the FMR1 gene – resulting in increased methylation of this region and silencing of the FMR1 gene (**epigenetics**)
- Characteristic features: Intellectual disability, learning difficulties, prominent ears, elongated face, macro orchidism (enlarged testis)
- Females are less severely affected than males (Only 50% of females with the full mutation may manifest with learning difficulties)
- Shows **anticipation** in successive generations (repeat expansion takes place during oogenesis; Disease tends to be more severe when a female transmits the mutation)
- Higher the number of repeats, greater is the severity of the disorder and earlier is the onset
- More details in MSK module



Fragile X syndrome: Diagnostic tests

- **Southern blot analysis** of the triplet repeats gives an indication of the number of triplet repeat sequences
- The X chromosomes show breakage (fragile X) in a folate deficient medium – Cytogenetic test. The fragile site is demonstrated only in individuals who carry the full mutation.
- Analysis of the mother's X chromosome usually demonstrates a premutation

